Unique Maximal Subgroup

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Group Theory Problems

Problem 1: Unique Maximal Subgroup

Let G be a finite group such that it has a unique maximal subgroup. Prove that G has prime power order.

The Proof

Part 1: Prove that G is cyclic.

Let H be the unique maximal subgroup of G. Since H is a proper subgroup, there exists some element $g \in G$ such that $g \notin H$. Consider the cyclic subgroup generated by g, denoted $\langle g \rangle$.

If $\langle g \rangle$ were a proper subgroup of G, it would have to be contained in *some* maximal subgroup. But since H is the *only* maximal subgroup, we would have $\langle g \rangle \subseteq H$, which immediately implies $g \in H$. This contradicts our initial assumption that $g \notin H$.

Therefore, $\langle g \rangle$ cannot be a proper subgroup. It must be that $\langle g \rangle = G$. Hence, G is cyclic.

Part 2: Prime Divisors of the Order of G

Now that we know G is cyclic, we can use the fundamental theorem of cyclic groups. If there exist two distinct prime divisors of the order of G, say p and q, then there must exist a unique subgroup of index p and a unique subgroup of index q.

Since subgroups of prime index are maximal, G would have at least two distinct maximal subgroups. But we are given that G has a unique maximal subgroup. Thus, it is impossible for the order of G to have more than one distinct prime divisor.

Conclusion: The order of G must be a prime power. ■